

PRODUCT SALES DASHBOARD – POWER BI PROJECT REPORT

This Power BI project analyzes sales, cost, and profit data to evaluate overall business performance.

Interactive dashboards and DAX measures are used to track key metrics such as total sales, profit, and profit margin.

The project enables category- and product-level analysis through dynamic visuals and filters. It supports data-driven decision-making by identifying high-performing and low-performing areas of the business.

1. Introduction

This Power BI project focuses on analyzing product sales performance using an interactive dashboard. The dashboard converts raw sales data into meaningful insights that help understand revenue generation, cost structure, profitability, and order distribution across products and categories. The project demonstrates the effective use of Power BI features such as DAX measures, interactive visuals, and filters to support data-driven decision-making.

The dashboard primarily analyzes two product categories: **Fruits** and **Vegetables**, and evaluates individual product performance within these categories.

2. Objectives of the Project

The main objectives of this Power BI project are:

- To analyze total sales, total cost, and total profit of products
- To compare product-wise and category-wise performance
- To identify high-profit and low-profit products
- To understand order distribution across products and categories
- To present insights using clear and interactive visualizations

3. Dataset Description

The dataset used in this project contains transactional sales data with the following columns:

- **Product** – Name of the product (Apple, Banana, Beans, etc.)

- **Category** – Product category (Fruit or Vegetables)
- **Order ID** – Unique identifier for each order
- **Sales** – Revenue generated from each product
- **Cost** – Cost incurred for selling the product
- **Profit** – Difference between sales and cost

The data was cleaned and validated to ensure consistency and accuracy before loading into Power BI.

4. Key Metrics and DAX Calculations

Key calculated measures were created using DAX to ensure dynamic calculations across the dashboard:

- **Total Sales** – Sum of all sales values
- **Total Cost** – Sum of all cost values
- **Total Profit** – Sum of profit values
- **Total Orders** – Count of distinct Order IDs
- **Profit Margin %** – Ratio of total profit to total sales

These measures automatically update based on filters applied in the dashboard, ensuring reliable analysis.

5. Dashboard Overview

The Product Sales Dashboard presents a comprehensive view of business performance through multiple visualizations:

- Product-wise sales and cost comparison
- Product-wise profit analysis
- Category-wise sales distribution
- Order count analysis by product and category
- Profit margin percentage by product

The layout is designed to provide both high-level summaries and detailed insights in a single view.

6. Sales and Cost Analysis

The **Total Sales and Total Cost by Product** visualization compares revenue and cost side by side for each product. This analysis helps identify products with high sales volumes and understand their associated costs.

Banana and Apple generate the highest sales among all products, while Mango and Beans contribute relatively lower sales. The comparison highlights how closely cost follows sales for most products, indicating consistent cost structures.

Additionally, **Total Sales by Category** shows that Fruits contribute a significantly higher share of total sales compared to Vegetables, making Fruits the dominant revenue-generating category.

7. Profit Analysis

The **Total Profit by Product** chart provides insights into the profitability of individual products. Banana is the most profitable product, followed by Apple and Broccoli. Products like Mango and Beans generate comparatively lower profit.

This analysis helps stakeholders identify products that contribute most to overall profit and those that may require cost optimization or pricing strategy review.

8. Profit Margin Analysis

The **Profit Margin % by Product** visualization focuses on efficiency rather than volume. It shows how much profit each product generates relative to its sales.

Products such as Banana, Carrots, and Apple show healthy profit margins, indicating efficient cost management. Products with lower margins highlight opportunities for improving operational efficiency.

9. Order Distribution Analysis

Order distribution is analyzed using **Count of Order ID by Category and Product** and **Count of Order ID by Product** visuals. Banana has the highest number of orders, indicating strong customer demand.

This analysis helps understand customer purchasing behavior and operational workload, as products with high order counts may require more inventory management and logistics support.

10. Insights and Business Impact

Key insights derived from the dashboard include:

- Fruits category is the top contributor to sales and profit
- Banana is the best-performing product in terms of sales, profit, and order count
- Some products show good sales but moderate profit margins, indicating potential pricing or cost improvements

The dashboard supports informed decision-making by clearly highlighting performance drivers and areas needing attention.

11. Conclusion

This Power BI project successfully demonstrates how interactive dashboards can be used to analyze sales performance, profitability, and customer demand. By combining DAX calculations with effective visualizations, the dashboard provides actionable insights that support strategic planning and operational improvements.

The project showcases practical Power BI skills and serves as a strong example of business intelligence reporting for academic, training, or professional use.